Electromagnetism

A Treatise on Electricity and Magnetism

IISER Bhopal · Saathvik Bogam

My Website

Contents

Chapter 1

Vector Calculus

"Then not Euclid, but elementary vectors, conjoined with algebra, and applied to geometry..."

Electro–Magnetic Theory, 1893
Oliver Heaviside

1.1 Grad, Div and Curl

We talk about importance of the gradient, the divergence and the curl, and how one applied them to scalar and vector fields.

1.1.1 Scalar and Vector Fields

A field is an object that prescribes a different object to every point in space. If the prescribed is a scalar, then it is a scalar field and takes the form of a function of (usually) three variables $\phi(x, y, z)$. If the prescribed object is a vector, then it is a vector field denoted by $\boldsymbol{v}(x, y, z)$. Each component of a vector field is an individual scalar field in the each direction.

$$\boldsymbol{v}(x, y, z) = v_x(x, y, z)\hat{\boldsymbol{x}} + v_y(x, y, z)\hat{\boldsymbol{y}} + v_z(x, y, z)\hat{\boldsymbol{z}} \tag{1.1}$$

1.1.2 Gradient

Definition 1.1 – Nabla Operator

One defines the Nabla/Del operator to be the following:

$$\nabla = \hat{\boldsymbol{x}}\frac{\partial}{\partial x} + \hat{\boldsymbol{y}}\frac{\partial}{\partial y} + \hat{\boldsymbol{z}}\frac{\partial}{\partial z} \tag{1.2}$$

We shall discuss how one applies this to scalar fields $\phi(x, y, z)$. The vector $\nabla\phi(\boldsymbol{r})$ is called the gradient of the scalar field ϕ at a point $\boldsymbol{r} = (x, y, z)$.

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\boldsymbol{x}} + \frac{\partial\phi}{\partial y}\hat{\boldsymbol{y}} + \frac{\partial\phi}{\partial z}\hat{\boldsymbol{z}} \tag{1.3}$$

To interpret what this means, we make use of the multivariate chain rule to write the differential of ϕ as:

$$d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \tag{1.4}$$

If we define a displacement vector $\boldsymbol{dl} = dx\hat{\boldsymbol{x}} + dy\hat{\boldsymbol{y}} + dz\hat{\boldsymbol{z}}$, then we can rewrite the differential of ϕ as:

$$d\phi = \nabla\phi \cdot \boldsymbol{dl} = |\nabla\phi||\boldsymbol{dl}|\cos\theta \tag{1.5}$$

If we fix the magnitude of $\boldsymbol{dl}$, then it turns out the maximum change in ϕ happens when the angle between $\boldsymbol{dl}$ and $\nabla\phi$ is zero, i.e. $\cos\theta = 1$. Therefore, we say that *the gradient points in the direction of maximum change, and its magnitude gives us the rate of increase in this maximal direction.*

If the gradient of a scalar field vanishes, then that point is a stationary point which could be a maximum (a summit), a minimum (a valley), a saddle point (a pass), or a “shoulder.

1.1.3 Divergence

The dot product of the nabla operator with a vector field $\boldsymbol{v}$ is called the divergence of that vector field.

$$\nabla \cdot \boldsymbol{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \tag{1.6}$$

The divergence of a vector about a point $\boldsymbol{r}$ gives the value of the ”spread” of the vector field. One way to imagine it is by considering the flow of water. If you drop some water and you see it spread out or diverge, then the divergence there is positive, since there is a spread of water. Using this, we define the terms ”faucet” or ”source” for a point with positive divergence and ”sink” or ”drain” for negative divergence. If a point has zero divergence, there is a equal amount of source and sink there, or no flow at all.

1.1.4 Curl

The cross product operation of nabla on a vector field is called the curl of the vector field,

$$\nabla \times \boldsymbol{v} = \sum_{i,j,k} \varepsilon_{ijk} \frac{\partial v_j}{\partial r_k}\hat{i} \tag{1.7}$$

where the index go in the order x, y, z and r_k is the k coordinate. It is reccomended to do cross products using the Levi–Civita symbol ε_{ijk} (please do this, it made my life doing cross product so much easier).

Definition 1.2 – Levi–Civita and Kronecker Delta

The Levi–Civita and Kronecker Delta symbols are tensors which are defined as follows:

$$\varepsilon_{ijk} = \begin{cases} 1, & \text{cyclic permutation } (xyz, yzx, zxy) \\ -1, & \text{anticyclic permutation } (zyx, yxz, xzy) \\ 0, & \text{otherwise} \end{cases} \tag{1.8}$$

$$\delta_{nm} = \begin{cases} 0, & m \neq n \\ 1, & m = n \end{cases} \tag{1.9}$$

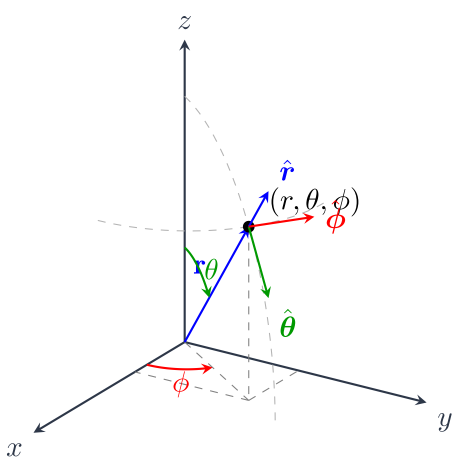
One should also note the identity:

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl} \tag{1.10}$$

The curl is a measure of how much and what direction the vector $\boldsymbol{v}$ swirls around the point in question. For example, a voxtex or a whirlpool is an example of a large magnitude of curl.

1.2 Curvilinear Coordinates

1.2.1 Spherical Coordinates



Let (x, y, z) be the position of a point in cartesian coordinates. Then we define (r, θ, ϕ) to be the position using spherical coordinates. r is the distance of the point from the origin, θ is the angle made with the $+z$–axis, and ϕ is the angle made by the projection on the x–y plane of the line joining the point and origin with the x–axis. Using the given diagram, one can discern that:

$$x = r \sin\theta \cos\phi \tag{1.11}$$

$$y = r \sin\theta \sin\phi \tag{1.12}$$

$$z = r \cos\phi \tag{1.13}$$

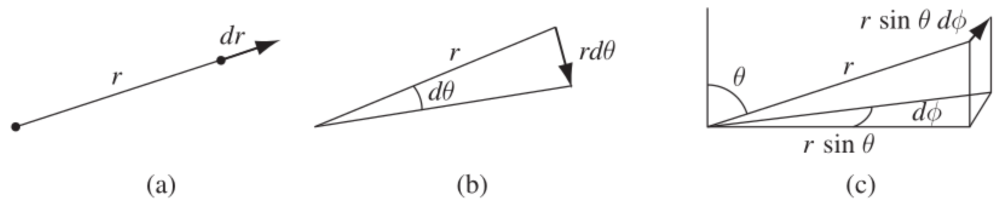
And the cartesian to curvilinear transformation being:

$$r = \sqrt{x^2 + y^2 + z^2} \in [0, \infty) \tag{1.14}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right) \in (0, \pi) \tag{1.15}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \in (0, 2\pi) \tag{1.16}$$

The unit vectors for the curvilinear coordinates are not fixed unlike the cartesian ones. This is a very important point to note, as the unit vectors will always be relative to the original point in space you are considering.



Now let's try to find the infinitesimal displacement vector $\boldsymbol{dl}$ for spherical coordinates. We do this by performing an infinitesimal transformation $(r, \theta, \phi) \to (r + dr, \theta + d\theta, \phi + d\phi)$ and then finding the displacement between the original and translated vectors. One can see that the radial part only increases in the radial direction, so $dl_r = dr$, but looking at the change in the $\hat{\boldsymbol{\theta}}$ direction, we see a displacement of $rd\theta$. Similarly, look at the $\hat{\boldsymbol{\phi}}$ direction, and we have a displacement of $r \sin\theta d\phi$. Therefore, we have the displacement vector to be:

$$d\boldsymbol{l} = dr\hat{\boldsymbol{r}} + rd\theta\hat{\boldsymbol{\theta}} + r\sin\theta\, d\phi\hat{\boldsymbol{\phi}} \tag{1.17}$$

We do not take it to be $d\boldsymbol{l} = dr\hat{\boldsymbol{r}} + d\theta\hat{\boldsymbol{\theta}} + d\phi\hat{\boldsymbol{\phi}}$ because this implies $d\theta$ and $d\phi$ have the units of length, which is simply not true since they are angles.

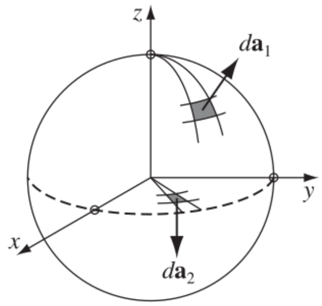
Another important quantity we consider is the infinitesimal volume area elements, $d\tau$ and $d\boldsymbol{a}$. The volume element is pretty simple, it is just the product of the components of the displacement vector, since it is small enough to be considered as a cuboid.

$$d\tau = dl_r dl_\theta dl_\phi = r^2 \sin\theta\, dr\, d\theta\, d\phi \tag{1.18}$$

The genereal formula depends on the surface, but we know the answer for two cases: one is if the area we are taking is a part of a sphere, then like the volume case, we consider it to small enough to take it as a rectangle of length and breadth dl_θ and dl_ϕ, with the normal in the $\hat{\boldsymbol{r}}$ direction:

$$d\boldsymbol{a}_1 = dl_\theta dl_\phi \hat{\boldsymbol{r}} = r^2 \sin\theta\, d\theta\, d\phi\, \hat{\boldsymbol{r}} \tag{1.19}$$

The second case is that of an area that lies on the x–y plane, and using the same technique,



we see its unit normal is the $\hat{\boldsymbol{\theta}}$ direction and the vector to be:

$$d\boldsymbol{a}_2 = dl_r dl_\phi \hat{\boldsymbol{\theta}} = r\,dr\,d\phi\,\hat{\boldsymbol{\theta}} \tag{1.20}$$

To obtain the nabla/del operations in spherical coordinates, you can just use the multi-variable chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial x} \tag{1.21}$$

and we substitute it into the operators to obtain:

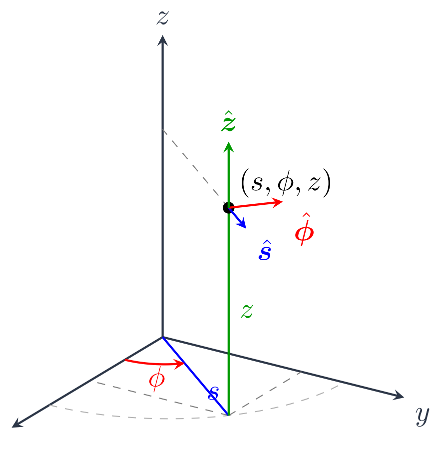
$$\nabla f = \frac{\partial f}{\partial r}\hat{\boldsymbol{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} \tag{1.22}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\, v_\theta) + \frac{1}{r\sin\theta}\frac{\partial v_\phi}{\partial \phi} \tag{1.23}$$

$$\nabla \times \boldsymbol{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta\, v_\phi) - \frac{\partial v_\theta}{\partial \phi}\right]\hat{\boldsymbol{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r}(r v_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(r v_\theta) - \frac{\partial v_r}{\partial \theta}\right]\hat{\boldsymbol{\phi}} \tag{1.24}$$

1.2.2 Cylindrical Coordinates

Let (x, y, z) represent a point in cartesian space, then the coordinates (s, ϕ, z) are the cylindrical coordinates, where s is the length of the projection of line joining the point and origin on the x–y plane, ϕ is the angle this projection makes with the x–axis (same as spherical) and z is the distance of the point from x–y plane (same as cartesian).



$$x = s\cos\phi \tag{1.25}$$

$$y = s\sin\phi \tag{1.26}$$

$$z = z \tag{1.27}$$

And the cartesian to cylindrical transformation is given by:

$$s = \sqrt{x^2 + y^2} \in [0, \infty) \tag{1.28}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \in (-\pi, \pi) \tag{1.29}$$

$$z = z \in (-\infty, \infty) \tag{1.30}$$

In this case, we have a fixed $\hat{\boldsymbol{z}}$ vector and variable $\hat{\boldsymbol{s}}$ and $\hat{\boldsymbol{\phi}}$ vectors. Now for a line element/infinitesimal displacement $d\boldsymbol{l}$, we have it to be:

$$d\boldsymbol{l} = ds\,\hat{\boldsymbol{s}} + s d\phi\,\hat{\boldsymbol{\phi}} + dz\,\hat{\boldsymbol{z}} \tag{1.31}$$

and the volume element is:

$$d\tau = s\,ds\,d\phi\,dz \tag{1.32}$$

The nabla/del operations in cylindrical coordinates is given as:

$$\nabla f = \frac{\partial f}{\partial s}\hat{\boldsymbol{s}} + \frac{1}{s}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z}\hat{\boldsymbol{z}} \tag{1.33}$$

$$\nabla \cdot \boldsymbol{v} = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z} \tag{1.34}$$

$$\nabla \times \boldsymbol{v} = \left(\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right)\hat{\boldsymbol{s}} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}\right)\hat{\boldsymbol{\phi}} + \frac{1}{s}\left(\frac{\partial}{\partial s}(s v_\phi) - \frac{\partial v_s}{\partial \phi}\right)\hat{\boldsymbol{z}} \tag{1.35}$$

1.3 Integrals

1.3.1 Line Integrals

A line integral is a way of integrating a vector valued function along a specific curve/path in space. If we wish to integrate over a path $\mathcal{P}$, the line integral is written as:

$$\int_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l}$$

If the path is a closed path, we write it as:

$$\oint_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l}$$

Essentially, we are taking the sum of components of the vector field that lies along this path $\mathcal{P}$. If we have a vector field that is independent of the path taken, then it is called a *conservative field*.

To illlustrate how one calculates a line integral, consider the following example:

Example 1.1 – Work done by a Force

Calculate the work done by the force $\boldsymbol{F} = y^2\hat{\boldsymbol{x}} + 2x\hat{\boldsymbol{y}}$ Newton in moving a body from $(1,-1)$ to $(1,1)$ over a parabolic path of $y = x^2$.

Solution: First, we know the work done by a force over a path is defined as the line integral of the force over that path.

$$W = \int_{\mathcal{P}} \boldsymbol{F} \cdot d\boldsymbol{l} = \int_{\mathcal{P}} (y^2\hat{\boldsymbol{x}} + 2x\hat{\boldsymbol{y}}) \cdot (dx\,\hat{\boldsymbol{x}} + dy\,\hat{\boldsymbol{y}}) \tag{1.36}$$

We now try to rewrite this in terms of x, by using the fact that $y = x^2 \implies dy = 2x\,dx$.

$$W = \int_{(-1,1)}^{(1,1)} y^2\,dx + 2x\,dy = \int_{-1}^{1} x^4 + 4x^2\,dx = \frac{46}{15}\text{ Joule} \tag{1.37}$$

1.3.2 Surface Integrals

Surface integrals are the sum of the component of the vector field lying in the direction of the surface area, which we take to be outward by convention. It represents what portion of the field actually passes through the surface $\mathcal{S}$.

$$\iint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a}$$

If we have a closed surface, we use the following symbol:

$$\oiint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a}$$

Example 1.2 – Electric Flux through a Surface

Consider a unit cube in the first octant with one of its corners at the origin, and an electric field $\boldsymbol{E} = 2xz\,\hat{\boldsymbol{x}} + 2y\,\hat{\boldsymbol{y}} + (xy + z)\,\hat{\boldsymbol{z}}$, now find the total flux through the cube.

Solution: The electric flux through a surface $\mathcal{S}$ is defined to be:

$$\Phi_E = \int_{\mathcal{S}} \boldsymbol{E} \cdot d\boldsymbol{a} \tag{1.38}$$

A cube has six sides, and taking one side at a time:

1. $x = 1$, $d\boldsymbol{a} = dy\,dz\,\hat{\boldsymbol{x}}$
$$\Phi_E = \int_0^1 \int_0^1 2z\,dy\,dz = 1 \tag{1.39}$$
2. $x = 0$, $d\boldsymbol{a} = -dy\,dz\,\hat{\boldsymbol{x}}$
$$\Phi_E = \int_0^1 \int_0^1 0\,dy\,dz = 0 \tag{1.40}$$
3. $y = 1$, $d\boldsymbol{a} = dx\,dz\,\hat{\boldsymbol{y}}$
$$\Phi_E = \int_0^1 \int_0^1 2\,dz\,dx = 2 \tag{1.41}$$
4. $y = 0$, $d\boldsymbol{a} = -dx\,dz\,\hat{\boldsymbol{y}}$
$$\Phi_E = \int_0^1 \int_0^1 0\,dx\,dz = 0 \tag{1.42}$$
5. $z = 1$, $d\boldsymbol{a} = dx\,dy\,\hat{\boldsymbol{x}}$
$$\Phi_E = \int_0^1 \int_0^1 xy + 1\,dx\,dy = 1.25 \tag{1.43}$$
6. $z = 0$, $d\boldsymbol{a} = -dx\,dy\,\hat{\boldsymbol{z}}$
$$\Phi_E = \int_0^1 \int_0^1 -xy\,dx\,dy = -0.25 \tag{1.44}$$

The total flux is the sum of all these, which is $\Phi_E = 4\ \mathrm{Nm^2C^{-1}}$

1.3.3 Volume Integrals

A volume integral is simply a sum of a quantity, for example density, over a volume to give mass. It is expressed as:

$$\iiint_{\mathcal{V}} \rho \, d\tau \tag{1.45}$$

where the cartesian volume element $d\tau = dx\,dy\,dz$. You can follow the given example to understand them better:

Example 1.3 – Mass of a Cube

A unit cube in the first octant with one of its corners at the origin has its density given by $\rho(x, y, z) = x^2 + 2zy$ kg m^{-3}, find its mass.

Solution: The mass is a simple volume integral of the density.

$$m = \int_{\mathcal{V}} \rho(x, y, z)\, d\tau = \int_0^1 \int_0^1 \int_0^1 x^2 + 2yz\, dx\, dy\, dz \tag{1.46}$$

$$= \int_0^1 \int_0^1 \frac{x^3}{3} + 2xyz\Big|_0^1 dy\, dz = \int_0^1 \int_0^1 \frac{1}{3} + 3yz\, dy\, dz \tag{1.47}$$

$$= \int_0^1 \frac{y}{3} + y^2 z\Big|_0^1 dz = \int_0^1 \frac{1}{3} + z\, dz \tag{1.48}$$

$$= \frac{z}{3} + \frac{z^2}{2}\Big|_0^1 = \frac{5}{6} \text{ kg} \tag{1.49}$$

1.4 Integral and Nabla Theorems

1.4.1 Gradient Theorem

Let $f(x, y, z)$ be a scalar function, and let $\boldsymbol{F} = \nabla f$. Now, we attempt to take the line integral of $\boldsymbol{F}$ over any path $\mathcal{P}$ from points $\boldsymbol{a}$ to $\boldsymbol{b}$.

$$\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{F} \cdot d\boldsymbol{l} = \int_{\boldsymbol{a}}^{\boldsymbol{b}} \nabla f \cdot d\boldsymbol{l} = \int_{\boldsymbol{a}}^{\boldsymbol{b}} \underbrace{\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz}_{df} \tag{1.50}$$

$$= \int_{\boldsymbol{a}}^{\boldsymbol{b}} df(x, y, z) = f(\boldsymbol{b}) - f(\boldsymbol{a}) \tag{1.51}$$

Essentially, we just proved that the line integral of the gradient of a function is always independent of the path chosen between two points. The function $f(x, y, z)$ is called the *scalar potential* of $\boldsymbol{F}$, and $\boldsymbol{F}$ is called a *conservative field*. Another property of conservative fields is that they have zero curl:

$$\nabla \times \boldsymbol{F} = \nabla \times (\nabla f) = 0 \tag{1.52}$$

1.4.2 Divergence Theorem

Consider a volume $\mathcal{V}$, which has a closed surface $\mathcal{S}$, then integral of a vector field $\boldsymbol{v}$ over that surface is equal to the volume integral of the divergence of $\boldsymbol{v}$.

$$\oiint_{\mathcal{S}} \boldsymbol{v} \cdot d\boldsymbol{a} = \iiint_{\mathcal{V}} \nabla \cdot \boldsymbol{v}\, d\tau \tag{1.53}$$

The formal proof for it is a bit difficult, so we shall not cover it. An interpretation of this theorem is if we consider the sources inside the volume, the surface integral gives us the actual flow that enters/exits, which should equate to the volume integral that gives us the sum of the spread of the flow.

An interesting thing one should note is the existence of divergence–less or *solenoidal* fields. This is true if and only if we have a vector field that is the curl of another i.e. $\boldsymbol{F} = \nabla \times \boldsymbol{A}$, and here we call $\boldsymbol{A}$ the *vector potential* of $\boldsymbol{F}$.

$$\nabla \cdot \boldsymbol{F} = \nabla \cdot (\nabla \times \boldsymbol{A}) = 0 \tag{1.54}$$

Vector potential is not unique, meaning we can add the gradient of any scalar function to $\boldsymbol{A}$ without changing the curl.

1.4.3 Stokes' Theorem

Take an open surface $\mathcal{S}$ which has a boundary, a closed path $\mathcal{P}$. Then, we have the following:

$$\oint_{\mathcal{P}} \boldsymbol{v} \cdot d\boldsymbol{l} = \iint_{\mathcal{S}} (\nabla \times \boldsymbol{v}) \cdot d\boldsymbol{a} \tag{1.55}$$

This is called *Stokes' Theorem*, and one may interpret this as the sum of the curls inside the surface is equal to the big curl around the boundary of that surface. For conservative forces, one can note that both sides are indeed zero, because the path integral is a difference of function at endpoints, and curl of a gradient is zero.

1.5 Dirac Delta Function

The Dirac Delta function is a particularly useful ”function”, which is defined as follows:

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases} \tag{1.56}$$

It can be interpreted as a large spike at $x = 0$, and as a limit of rectangle of unit area. This unit area part is important because we also define, for any $\varepsilon > 0$:

$$\int_{-\varepsilon}^{\varepsilon} \delta(x)\, dx = 1 \tag{1.57}$$

Any integral that contains the origin will give an area of 1. Since $\delta(x)$ is zero everywhere except the origin, then $f(x)\delta(x) = f(0)\delta(x)$, or for a shifted delta, we have $f(x)\delta(x-a) = f(a)\delta(x-a)$.

$$\int_{a-\varepsilon}^{a+\varepsilon} f(x)\delta(x-a)\,dx = f(a) \tag{1.58}$$

We similarly define a 3D version of it, which we write as:

$$\delta^3(\boldsymbol{\mathscr{r}}) = \delta(x-x')\delta(y-y')\delta(z-z') \tag{1.59}$$

Here, $\boldsymbol{\mathscr{r}} = \boldsymbol{r} - \boldsymbol{r}'$, where $\boldsymbol{r} = (x, y, z)$ and $\boldsymbol{r}' = (x', y', z')$. And now generalizing equation 1.58, we have:

$$\iiint_{\boldsymbol{r}'-\varepsilon}^{\boldsymbol{r}'+\varepsilon} f(\boldsymbol{r})\delta^3(\boldsymbol{\mathscr{r}})\,dx\,dy\,dz = f(\boldsymbol{r}') \tag{1.60}$$

Using the properties of the delta function, we find the gradient and divergence of the following fields as follows:

$$\nabla \cdot \left(\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi\delta^3(\boldsymbol{\mathscr{r}}) \tag{1.61}$$

$$\nabla^2 \left(\frac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}}) \tag{1.62}$$